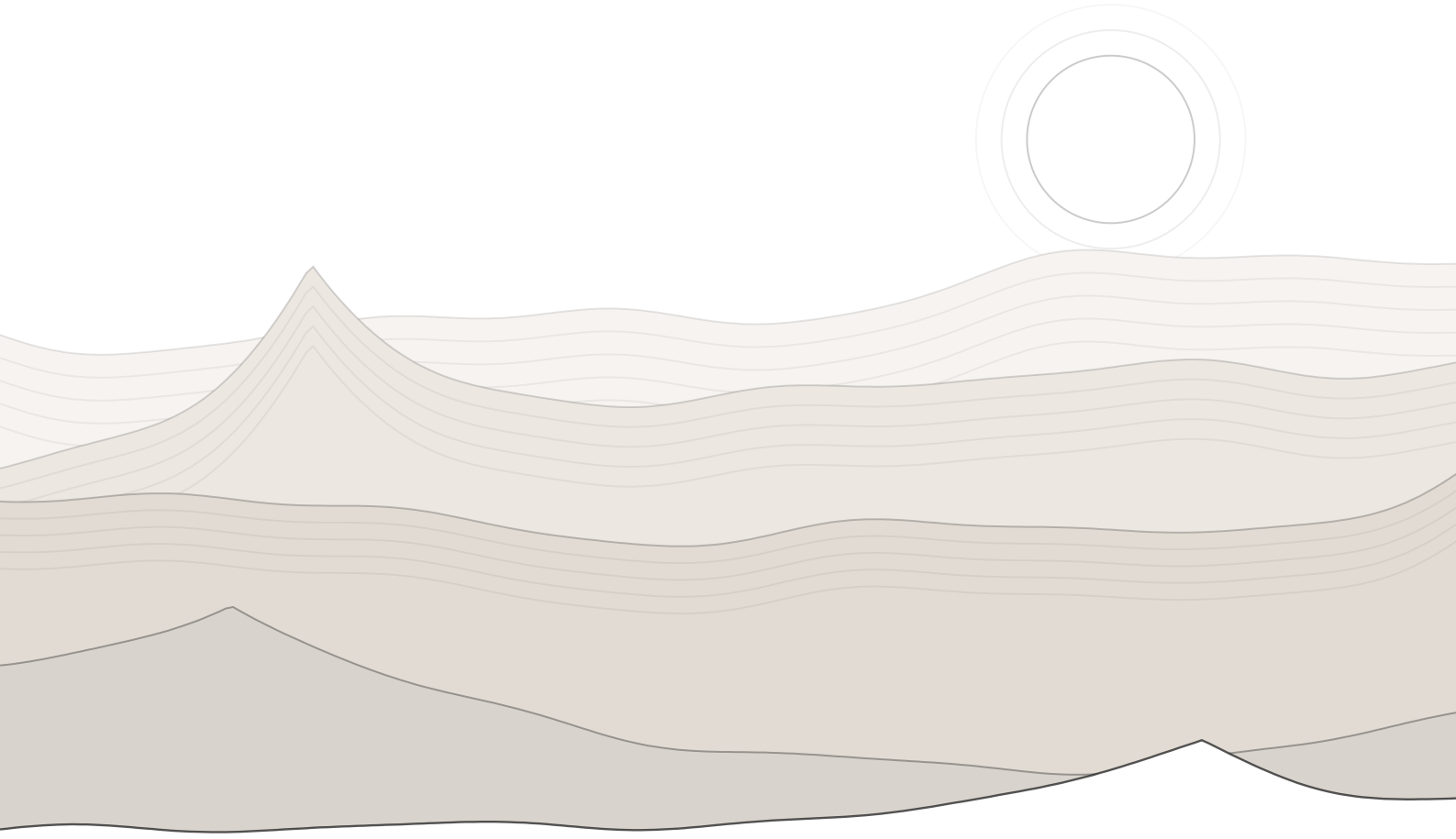
AMC 2026 · ROUND 1 · GROUP 3

Strategic Asset Allocation

Part A, the analysis of the allocation: outlook, universe, inputs, method, allocation, backtest, code and portfolio. Part B, the checkpoint assignments: investment policy (page 2), robust approaches and backtest protocol.



STARTING NAV	VOLATILITY TARGET	PUBLISHED
USD 100M	8% p.a.	16.10.26

Investment policy

1 · UNIVERSE

Long-only ETFs that track broad asset classes, listed in the US and traded in USD. No single stocks, no short positions, no leverage, few positions; cash earns no interest. Eight strategic ETFs in three baskets, one per macro regime, plus a T-bill ETF used only as the cash leg (sections 02–03).

EQUITY · GROWTH	BONDS · DISINFLATION	REAL ASSETS · INFLATION	CASH LEG
VTI · VEA · IEMG	BND · BNDX · TIP	GLD · BCI	BIL

2 · OBJECTIVE AND RISK BUDGET

Objective: the highest Sharpe Ratio over the 1-month T-bill, with an allocation that performs acceptably in each of the four macro regimes, since we cannot predict which one comes next (section 01).

VOL. TARGET	EX-ANTE VOL.	1-MONTH 5% VAR	MAX DRAWDOWN
8%	8.13%	3.86%	−15%
Range 7–9% p.a.	Last 60 months	= USD 3.86 million	Acceptable · backtest −13.0%

The worst loss of our weights since 2013 was −13.0%, in 2022, when stocks and bonds fell together. We accept about twice the target volatility; a deeper fall would mean the allocation behaves differently from what we tested, and we would review it at Round 2.

3 · CONSTRAINTS, WITH REASONS

CONSTRAINT WE IMPOSE	REASON (TODAY'S MODEL ERRORS)
Concentration. Fixed risk budgets: equity 50%, bonds 25%, real assets 25%, equal risk per ETF inside each. No ETF above 20% of the capital or of the risk.	Estimation error: five years of returns give a standard error of about 7 pp on the mean of a 15%-vol ETF, so expected returns cannot set the weights. An optimiser on them would hold 35% in gold, the recent winner. Risk budgets need only the covariance matrix.
Regions. Equity risk split equally across US, developed ex-US and emerging markets (35 / 33 / 33 of the equity capital, against 64% US in the MSCI ACWI [6]). Non-US bonds hedged to USD.	The last five years favour the US (historical excess return 9.0% against 5.9% for emerging markets), but a past mean is not a forecast, and market-cap weights would bet the equity risk on one economy and a few mega caps. Currency risk in bonds is not rewarded.
Liquidity. Only large, liquid index ETFs; whole shares, rounded down; no position below 2% of the capital (it would go to BIL).	Model error: mean–variance assumes every position trades at the closing price at no cost. Broad ETFs make that assumption close to true, and keep the portfolio easy to adjust in Rounds 2–3.
Cash. No idle cash: BIL is the only cash leg, equal to 1 − the risky weights, and never an input of the optimiser.	Cash earns 0 in the challenge, about −4% a year in excess of the T-bill (4.13%). BIL earns the T-bill rate, so its excess return is close to zero.

4 · RISK CONTRIBUTIONS

ETF	WEIGHT	RISK CONTR.	ETF	WEIGHT	RISK CONTR.
VTI	10.90%	15.21%	TIP	15.26%	8.84%
VEA	10.24%	17.42%	GLD	11.15%	15.49%
IEMG	10.23%	17.36%	BCI	12.40%	9.51%
BND	13.33%	8.34%	BIL	0.01%	0.00%
BNDX	16.48%	7.83%	Total	100.00%	100.00%

Position above half of the risk: none. The largest single contribution is VEA, 17.4%. By basket, equity holds 31.4% of the capital and exactly 50.0% of the risk; bonds 45.1% and 25.0%; real assets 23.5% and 25.0%. Half of the risk in equity is intended: it is the budget we give to the equity premium, the best-documented long-run premium, and far from the 90% of a 60/40 portfolio (Class 3).

Market outlook

Our reading of the macro picture is that 2026 has been a year of stagflation, that economists expect a return to goldilocks in 2027, and that markets do not share this optimism. Fed funds futures, in fact, price about 3.4 further hikes, reaching roughly 100 basis points above what economists forecast. Hence, we do not build the portfolio around a single regime, but hold assets that perform in each of them.

1.1 · GROWTH AND INFLATION

We classify each economy into the four regimes defined by the direction of growth and inflation, using the Bloomberg consensus forecasts. During 2026 inflation rose in every region, in the United States and in the emerging economies it did so while growth was slowing: US CPI went from 2.7% to 3.4% and real GDP growth fell from 2.6% to 1.8%, which places both economies in stagflation. For 2027, the consensus expects growth to remain steady or to rise and inflation to fall in every region, the pattern of a goldilocks economy. [1]

TABLE 1 · CONSENSUS FORECASTS AND MACRO REGIMES

REGION	GDP Q4 26	GDP Q4 27	CPI Q4 26	CPI Q4 27	REGIME 2026
United States	2.1%	2.1%	3.5%	2.1%	Stagflation
Developed economies	1.8%	1.9%	3.9%	2.5%	Overheated
Emerging economies	3.9%	4.4%	3.6%	2.9%	Stagflation

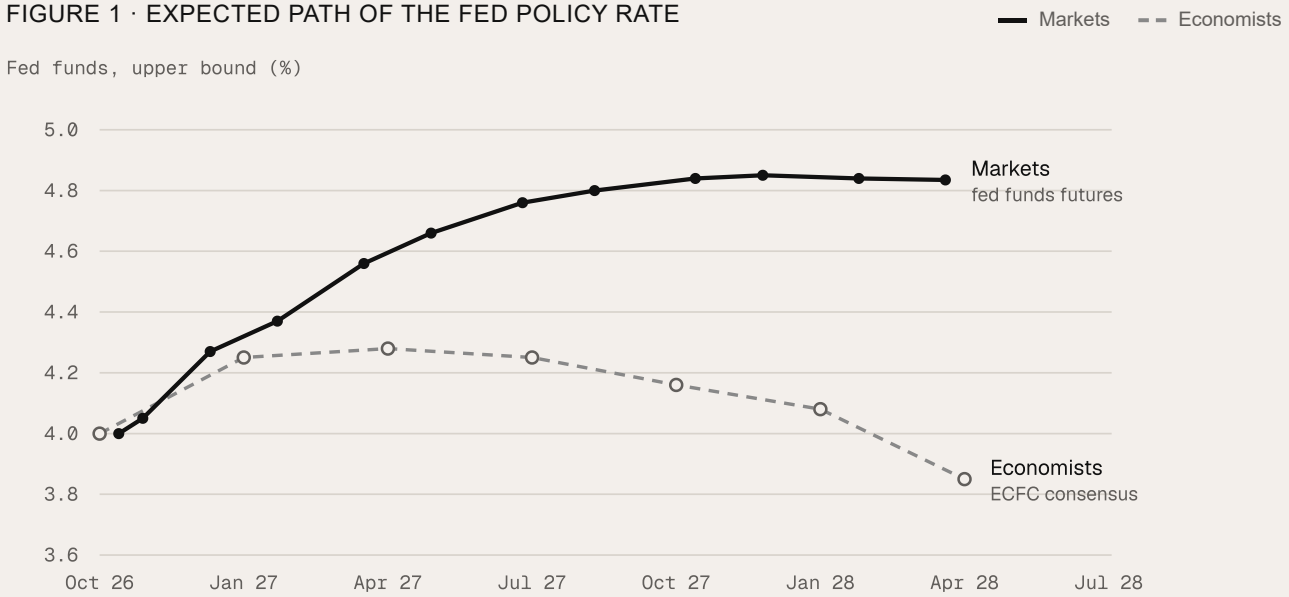
Real GDP and CPI, year-on-year. Regime 2026: direction of growth and inflation from Q1 to Q3 2026. [1]

Behind these numbers, the US economy is slowing gently rather than contracting. Unemployment is stable at about 4.2% and payroll growth has eased to about 60,000 jobs a month, while private investment remains the strongest component of demand, having grown at an annualised 7.4% in the third quarter. Inflation is still the main concern, with core PCE at 3.3%. Europe continues to grow slowly, at 1.3–1.6%, whereas the emerging economies combine rising growth with monetary easing, as their central banks are expected to cut rates from 6.3% to about 5.0% by early 2028. [1]

Monetary policy: economists versus markets

After the September hike the Fed funds target range stands at 3.75–4.00%, and the path of the policy rate is where economists and markets disagree most. Economists expect one further hike followed by cuts that would bring the rate to 3.85% by early 2028, while fed funds futures price about 3.4 additional hikes, to an upper bound of about 4.84% by the end of 2027. [1, 2]

FIGURE 1 · EXPECTED PATH OF THE FED POLICY RATE



Markets: implied upper bound from fed funds futures. Economists: Bloomberg ECFC consensus. [1, 2]

IF ECONOMISTS ARE RIGHT

Bonds rally and growth assets benefit from lower rates.

IF MARKETS ARE RIGHT

Inflation remains sticky and both bonds and equities come under pressure.

We read this gap as the clearest evidence that the next regime is genuinely uncertain, and it is the starting point of our allocation.

1.3 · RATES, INFLATION PRICING, CREDIT AND THE DOLLAR

INDICATOR	1 YEAR AGO	LATEST
US Treasury 1-month yield	4.20%	4.13%
US Treasury 2-year yield	3.60%	4.80%
US Treasury 10-year yield	4.14%	5.24%
US Treasury 30-year yield	4.72%	5.60%
5-year breakeven inflation	2.39%*	2.34%
10-year real yield (TIPS)	1.78%	2.87%
US investment-grade spread (OAS)	0.75%	0.82%
US high-yield spread (OAS)	2.84%	3.15%
VIX	16.3	15.4
EUR/USD, consensus Q3 2026 → Q4 2027	1.13	1.17

*Table 2 · Key market indicators. Treasury yields and 1-year-ago real yield (8 Oct 2025): U.S. Treasury par and real yield curves [3]. Latest breakeven, real yield, spreads and VIX: FRED (T5YIE, DFII10, BAMLC0A0CM, BAMLH0A0HYM2, VIXCLS) [4]. *1-year-ago breakeven (9 Oct 2025) calculated as 5Y nominal 3.74% − 5Y real 1.35%. 1-year-ago VIX: close of 8 Oct 2025 [12]. EUR/USD [1].*

The indicators in Table 2 point to four features of the current market that shape our universe. The Treasury curve has shifted up by 80–130 basis points from the one-year maturity onwards, so that bonds once again offer a meaningful income. The inflation priced by the market, 2.34% over five years, is lower than the consensus forecast of about 2.8% for the next four quarters, which makes inflation protection inexpensive. Over the year the 10-year real yield rose by about 109 bp (1.78% → 2.87%) while five-year inflation expectations were almost unchanged (2.39% → 2.34%): the rise in nominal yields came from real rates, not from inflation expectations. Credit spreads remain tight, so that credit offers little compensation for its risk, and the consensus expects a mild weakening of the dollar, which would support non-US assets measured in USD.

How we built the universe

01

No bet on a single regime

Given the outlook above, we hold assets that perform in each of the four regimes rather than picking one.

02

Cheap, broad market exposures

The policy portfolio drives most of the risk and return, and market beta should be bought passively. Every ETF tracks a broad market-cap index at low cost. Sector, factor and thematic ETFs are left for the tactical rounds, where they are measured as deviations.

03

No overlaps

We did not select ETFs that cover the same universe.

EQUITY	FIXED INCOME	REAL ASSETS	RISK TOOL
VTI · VEA · IEMG	BND · BNDX · TIP	GLD · BCI	BIL

Eight strategic ETFs — three equity, three fixed income, two real assets — plus one T-bill ETF used only as a risk-management tool.

The ETFs and why we chose them

3.1 · EQUITY

Equity carries the equity risk premium, the most reliable source of long-run excess return. Instead of a single world ETF we hold three regional blocks: this gives us control over the regional mix and makes our outlook visible in the portfolio.

VTI Vanguard Total Stock Market ETF

US equity · CRSP US Total Market Index · listed May 2001 · 3,531 holdings · TER 0.03% (30 Jun 2026) [5]

WHY WE CHOSE IT	The US is the deepest and most innovative equity market: it represents 64% of the MSCI ACWI [6]. US companies have delivered the strongest earnings growth of the last decade, driven by technology and, more recently, the investment cycle in artificial intelligence, visible in the strength of private investment. US equity is therefore the main source of expected return in our portfolio.
WHY NOT SPY / VOO	SPY and VOO hold the 500 largest companies; VTI holds about 3,500, adding mid and small caps that are more sensitive to domestic growth and to falling rates, and slightly less concentrated in the largest technology names.
RISKS WE ACCEPT	High valuations and concentration in mega caps; with the 10-year yield at 5.24% bonds are a serious alternative, which compresses the equity risk premium. We limit these risks with the concentration cap and by holding equity outside the US.

VEA Vanguard FTSE Developed Markets ETF

Developed ex-US equity · FTSE Developed All Cap ex US Index · listed July 2007

WHY WE CHOSE IT	We do not want the portfolio to depend on one economy. Developed ex-US markets (Europe, Japan, UK, Canada, Australia, South Korea) trade at a valuation discount to the US and have a different sector mix — more financials, industrials and healthcare, less technology — so they respond differently to the same shocks. The fund is unhedged: the expected weakening of the dollar (EUR/USD +3.5% to end-2027) would add to its returns in USD.
WHY NOT EFA / VXUS	EFA excludes Canada and small caps; VEA includes both. We preferred VEA to an all-in-one ex-US fund such as VXUS to size emerging markets separately.
RISKS WE ACCEPT	With European growth expected at only 1.3–1.6%, the case for this position rests on valuations rather than on earnings momentum, and it may take time to pay off. Many European and Japanese companies are large exporters, which leaves them exposed to tariffs and trade tensions, and since the fund is unhedged, a stronger dollar would reduce its value in USD.

IEMG iShares Core MSCI Emerging Markets ETF

Emerging markets equity · MSCI Emerging Markets Investable Market Index · listed October 2012

WHY WE CHOSE IT

Emerging markets offer the best macro combination of the three regions: growth rising from 3.9% to 4.4% in 2027, inflation falling and central banks easing by about 140 bp [1], with current-account surpluses of about 3% of GDP that reduce their dependence on foreign capital. They also offer lower valuations and access to the semiconductor supply chain (Taiwan, Korea). A separate position lets us hold them at least at market weight.

WHY NOT EEM

IEMG includes small caps (IMI index) and is cheaper.

RISKS WE ACCEPT

Emerging markets are more volatile than developed ones and carry political and regulatory risk, especially in China, where government intervention can change the outlook for entire sectors. They also tend to suffer when the dollar strengthens and US rates rise, as capital flows back to the United States. Finally, the position partly overlaps with VEA: South Korea is classified as Developed by FTSE since 2009 [7] but remained Emerging for MSCI in June 2026 [8], so it is held in both funds.

3.2 · FIXED INCOME

Bonds lower the volatility of the portfolio, which is necessary to stay within the 7–9% risk budget, and protect it in a recession with falling inflation. We hold three different sources of fixed-income return: US nominal rates, non-US rates and US real rates.

BND Vanguard Total Bond Market ETF

US investment-grade bonds · Bloomberg US Aggregate Float Adjusted Index · listed April 2007

WHY WE CHOSE IT

BND holds the US investment-grade market: Treasuries, agency mortgage-backed securities and IG corporate bonds, with an intermediate duration. With the 10-year yield at 5.24% and the 2-year at 4.80%, bonds pay a meaningful income; economists expect the 10-year yield to fall to about 4.6% by end-2027 [1], which would add a capital gain. Above all, BND is our insurance against a recession: when growth and inflation fall, central banks cut and bond prices rise.

WHY NOT TLT

The US budget deficit is expected at about 6.5% of GDP and public debt at 105% of GDP [1]: a persistent supply of Treasuries may keep long-term yields high. Intermediate duration captures most of the recession hedge with much less of this risk. We excluded high-yield bonds: with a spread of 3.15% they pay little for a credit risk that rises exactly when equities fall.

RISKS WE ACCEPT

Since 2021 the stock–bond correlation has turned positive: when inflation is the problem, bonds and equities can fall together, as in 2022. Futures also price further Fed hikes.

BNDX Vanguard Total International Bond ETF

International IG bonds, USD-hedged · Bloomberg Global Aggregate ex-USD Float Adjusted RIC Capped Index · listed June 2013

WHY WE CHOSE IT	Rate cycles outside the US are not synchronised with the Fed: the ECB rate is about 2.85% and expected to ease to 2.5% by early 2028, and the euro-area 10-year yield is expected to fall from 3.9% to 3.7% [1]. BNDX therefore diversifies the duration risk of BND. Because it is hedged, it behaves like a bond and not like a currency position; with US short rates above European and Japanese rates, the hedge also earns the interest-rate differential.
WHY NOT IGOV	IGOV is unhedged: currency moves would add equity-like volatility to the defensive part of the portfolio.
RISKS WE ACCEPT	A global inflation shock would push yields up everywhere at once.

TIP iShares TIPS Bond ETF

US inflation-linked Treasuries · listed December 2003

WHY WE CHOSE IT	Inflation hurts stocks and bonds at the same time, and 2026 has been a year of rising inflation. TIP is the most direct protection, and today it is cheap: the 5-year breakeven (2.34%) is below the consensus CPI for the next four quarters (2.8%) and in line with the consensus for 2028 (2.3%) [1, 4]. The real yield of 2.87% — up about 109 bp in a year while breakevens barely moved — means TIP pays a solid return even if inflation simply returns to target.
WHY NOT STIP	STIP holds only 0–5 year maturities. We preferred TIP because, with real yields at their highest in years, we want to lock in some real-rate duration.
RISKS WE ACCEPT	TIPS protect against inflation, not against rising real yields: if real rates keep increasing, TIP can lose value even while inflation is high, as happened in 2022. Conversely, if inflation falls back faster than the market expects, the inflation adjustment shrinks and TIP is likely to underperform nominal bonds such as BND.

3.3 · REAL ASSETS

Real assets protect the portfolio in the two regimes in which equities and nominal bonds can fall together — high inflation and supply shocks — and are the least correlated part of the universe.

GLD SPDR Gold Shares

Gold · physical gold bullion · listed November 2004

WHY WE CHOSE IT	Gold has no credit or counterparty risk and has historically held its value in stagflation, geopolitical crises and periods of dollar weakness. Official demand has been a structural support: central banks bought about 850 tonnes in 2025 [9]. With an uncertain outlook, high US public debt and an expected weaker dollar, we consider gold a necessary insurance.
WHY NOT IAU	Both hold physical gold; GLD is the largest and most liquid gold ETF and has the longest history. IAU is cheaper, but for a position held a few months liquidity matters more than fees.
RISKS WE ACCEPT	Gold pays no income, and with real yields near 2.9% the opportunity cost is high. It is volatile and has rallied strongly: the LBMA price averaged a record USD 4,873/oz in Q1 2026, and central-bank buying slowed sharply in that quarter (revised to 57 tonnes, with visible selling) [9].

BCI Bloomberg All Commodity Strategy K-1 Free ETF

Broad commodities · Bloomberg Commodity Index Total Return · listed March 2017

WHY WE CHOSE IT	Commodities perform best when inflation comes from supply shocks — energy disruptions, geopolitical conflicts, bad harvests — the scenario in which both stocks and bonds lose, as in 2022. They complement gold: gold reacts to monetary and geopolitical stress, commodities to the real economy. The total-return index also earns the T-bill rate on the collateral.
WHY NOT DBC	BCI follows the Bloomberg Commodity Index: 23 commodities, with no sector above 33% and no single commodity above 15% [11]. DBC is about half energy and more expensive [10].
RISKS WE ACCEPT	Roll costs in markets in contango and sharp falls in a recession, when demand for raw materials collapses.

BIL SPDR Bloomberg 1-3 Month T-Bill ETF

3.4 Risk-management tool · US Treasury bills, 1-3 months · listed May 2007

ROLE	BIL is not a strategic position but a tool to manage the risk budget. If the eight strategic ETFs together have an ex-ante volatility above 9%, we scale the risky portfolio down to the 8% target and invest the remainder in BIL; if the volatility is already within the 7–9% band, BIL is not held at all. Since T-bills barely move, BIL can only bring the volatility of the portfolio down, never raise it.
WHY NOT CASH	The Sharpe Ratio is measured on returns in excess of the 1-month T-bill (4.13%) [3]. Cash earns nothing, so it has an excess return of about −4% a year and lowers the Sharpe Ratio; BIL earns about the T-bill rate, so its excess return is close to zero.
WHY NOT SGOV	SGOV is an equivalent and slightly cheaper alternative; we use BIL because its longer track record (2007, against 2020 for SGOV) gives more data if the backtest sample is extended.

The inputs

All inputs are estimated on the same 60 complete months, October 2021 – September 2026, as the guidelines require. Returns are monthly total returns in USD, computed from Bloomberg adjusted prices, which include dividends [13]. Means and variances are annualised ×12, volatilities ×√12. BCI only started trading in 2017, so for the whole sample we use the index it tracks, the Bloomberg Commodity Index Total Return (BCOMTR), before the fund's fees. In this way, the estimation window and the backtest use the same series. The risk-free rate is the 1-month T-bill (FRED DGS1MO) [14], the rate on which the Sharpe Ratio of the challenge is computed.

TABLE 3 · VOLATILITY AND MEAN

ETF	VOLATILITY	HIST. EXCESS RETURN*
VTI	15.24%	8.97%
VEA	15.67%	7.02%
IEMG	16.28%	5.86%
BND	6.76%	−4.25%
BNDX	5.57%	−3.74%
TIP	6.00%	−3.87%
GLD	17.08%	14.58%
BCI†	15.36%	8.10%

**Mean monthly return minus the T-bill, ×12. Shown, not used by our method. †BCOMTR index.*

TABLE 4 · CORRELATIONS

	VTI	VEA	IEMG	BND	BNDX	TIP	GLD	BCI
VTI	1	0.82	0.72	0.50	0.53	0.53	0.14	0.09
VEA	0.82	1	0.81	0.64	0.58	0.62	0.42	0.20
IEMG	0.72	0.81	1	0.56	0.47	0.53	0.44	0.24
BND	0.50	0.64	0.56	1	0.89	0.90	0.46	−0.06
BNDX	0.53	0.58	0.47	0.89	1	0.85	0.41	−0.13
TIP	0.53	0.62	0.53	0.90	0.85	1	0.47	0.12
GLD	0.14	0.42	0.44	0.46	0.41	0.47	1	0.31
BCI	0.09	0.20	0.24	−0.06	−0.13	0.12	0.31	1

Shading: dark ≥ 0.80, light 0.50–0.79. Monthly returns, Oct 2021 – Sep 2026.

4.1 · WHAT THE CORRELATIONS TELL US

Each basket behaves almost as one asset. The three equity ETFs correlate 0.72–0.82 and the three bond ETFs 0.85–0.90, so there is little diversification inside a basket. Diversification comes from combining the baskets, which is why our method sets the risk of each basket (section 05).

Stocks and bonds have moved together. Their correlation is 0.47–0.64 over a window that includes 2022. Bonds still lower the volatility, but they hedge equities less than they did before 2021 (Class 3, Part II). For this reason we do not rely on bonds alone for protection.

Gold and commodities are the real diversifiers. Their correlations with the other ETFs are at most 0.47, and commodities are uncorrelated with bonds (−0.13 to 0.12). This is the statistical side of the role that we give to real assets in section 03.

4.2 · WHY WE DO NOT USE THE EXPECTED RETURNS

The historical excess returns describe the last five years, not the next ones. Bonds show −4% a year because yields rose sharply in 2022, the same rise that now gives bonds a yield above 5%. Gold shows +14.6% because of a record rally. With five years of data, the standard error of the mean of an ETF with a 15% volatility is 15% / √5 ≈ 6.7 percentage points, larger than most of the means themselves. An optimiser using these numbers puts a third of the capital in gold and nothing in bonds (section 05). Our weights therefore depend only on the volatilities and the correlations, which are estimated much more precisely.

The method: risk budgets by basket

Our weights come from a two-level risk-budgeting rule that uses only the covariance matrix. Inside each basket every ETF carries the same share of the basket's risk (equal risk contribution, ERC). Across the baskets the risk is split 50% equity, 25% bonds, 25% real assets. The rules of the round apply: long-only, no leverage, ex-ante volatility between 7% and 9%.

5.1 · HOW THE WEIGHTS ARE COMPUTED

Step 1 · inside the basket

ERC weights, so that $w_i(Vw)_i / w'Vw$ is the same for every ETF of the basket. Unlike 1/σ, ERC uses the correlations.

Step 2 · across baskets

Each basket becomes one return series. On their covariance matrix we solve min $\Sigma_k(\%RC_k - b_k)^2$, with b = 50 / 25 / 25, w ≥ 0, Σw ≤ 1 and 7% ≤ σ ≤ 9%.

Step 3 · combine

ETF weight = basket weight × weight inside the basket; whatever is not invested goes to BIL. Solved numerically (SLSQP, scipy).

5.2 · WHY A RISK-BASED METHOD

Estimation error. Mean–variance turns small errors in expected returns into extreme weights. In the 40-year horse race of Class 3, its out-of-sample Sharpe Ratio fell to 0.04, against 0.53 for risk parity. Our inputs (section 04) show the same problem: historical means are both noisy and backward-looking. A rule that needs no expected returns avoids it.

Regimes. Equities pay in goldilocks, bonds in a recession with falling inflation, real assets under inflation (Class 3). Our three baskets map onto these regimes, and fixing their risk shares means that no single regime can dominate the result. This fits our outlook: the next regime is uncertain (section 01).

Capital is not risk. A 60/40 portfolio has about 90% of its risk in equities. By budgeting the risk directly we control what each basket can do to the portfolio, whatever its capital weight.

5.3 · WHY 50 / 25 / 25

Equal risk across the baskets would be the maximum-Sharpe portfolio only if the baskets had the same Sharpe Ratio and the same correlations [15]. Long-run evidence says they do not: the equity premium is the largest and best-documented premium, and it is backed by cash flows [16], while gold has none and the commodity premium is debated. We therefore split the risk 50/50 between growth and protection, and the protection equally between bonds and real assets. This is a prior on the ranking of the premia, not an estimated return.

It is also the budget closest to the 8% target. Out of sample, the four budgets we tested have practically the same Sharpe Ratio (Table 5), so a larger equity budget would add volatility but no risk-adjusted return.

TABLE 5 · THE FOUR EQUITY BUDGETS

RISK BUDGET EQUITY / BONDS / REAL	EX-ANTE VOL.	BIL	SHARPE, HOLD-OUT	SHARPE, WALK-FORWARD
33 / 33 / 33	7.39%	0.06%	0.65	0.47
50 / 25 / 25 · our SAA	**8.13%**	**0.01%**	**0.66**	**0.48**
60 / 20 / 20	8.71%	0.00%	0.66	0.48
70 / 15 / 15	9.00%	4.71%	0.66	0.49

Hold-out: Jul 2013 – Sep 2021, months not used for the weights. Walk-forward: Jul 2018 – Sep 2026, weights re-estimated every month on the previous 60 (section 07). At 70/15/15 the 9% cap binds and the rest goes to BIL.

The methods we compared

Before choosing, we computed eleven allocations on the same inputs and under the same rules (long-only, no leverage, rest in BIL, ex-ante volatility in 7–9%). The comparison shows what each rule does with our universe, and where a rule depends on an estimated expected return.

TABLE 6 · WEIGHTS OF THE ELEVEN ALLOCATIONS

METHOD	VTI	VEA	IEMG	BND	BNDX	TIP	GLD	BCI	BIL	VOL.
Equal weight	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	0.0	8.78
Risk parity 1/σ	8.1	7.9	7.6	18.3	22.2	20.6	7.2	8.1	0.0	7.23
Minimum variance	9.8	9.2	9.2	11.3	11.6	11.4	9.8	11.0	16.7	7.00
Maximum Sharpe	26.0	0.0	0.0	0.0	0.0	0.0	35.3	14.6	24.0	8.61
Mean–variance at 8%	24.0	0.0	0.0	0.0	0.0	0.0	32.9	13.5	29.5	8.00
ERC, 8 ETFs	8.2	6.7	6.8	16.7	21.8	17.9	8.0	13.7	0.2	7.13
Max. diversification	16.4	0.0	0.0	5.3	41.4	0.0	10.6	26.4	0.0	7.00
Basket 33/33/33	7.2	6.7	6.7	15.7	19.4	18.0	12.4	13.8	0.1	7.39
Basket 50/25/25 · SAA	**10.9**	**10.2**	**10.2**	**13.3**	**16.5**	**15.3**	**11.1**	**12.4**	**0.0**	**8.13**
Basket 60/20/20	13.5	12.7	12.7	11.7	14.5	13.4	10.2	11.4	0.0	8.71
Basket 70/15/15	15.8	14.8	14.8	9.4	11.6	10.8	8.6	9.5	4.7	9.00

Weights and ex-ante volatility in %, Oct 2021 – Sep 2026. Equal weight and 1/σ are fully invested by definition; the others are solved under the 7–9% band.

WHY WE DID NOT CHOOSE THE OTHERS

Maximum Sharpe, mean–variance	They are the only rules that use the expected returns, and it shows: a third of the capital in gold, a quarter in US equity, no bonds and no non-US equity, and 24–30% in BIL. They load on the winners of the last five years, which is the estimation-error problem of Class 3.
Minimum variance	It stops at the 7% floor of the band, and the solution is not unique: every portfolio with exactly 7% volatility is optimal, so the weights depend on where the solver starts. It also leaves 16.7% in BIL.
Maximum diversification	It concentrates 41% in BNDX and 26% in commodities and drops non-US equity: a portfolio built on one estimated correlation structure.
ERC on the 8 ETFs, basket 33/33/33	Both stay near the bottom of the band (7.1% and 7.4%). In flat ERC the number of ETFs per asset class becomes a hidden risk budget: three bond ETFs out of eight receive 3/8 of the risk. Equal baskets give equity only a third of the risk.
Equal weight, risk parity 1/σ	These are the robust benchmarks of Checkpoint 2. Equal weight ignores the risk of each ETF; 1/σ ignores the correlations.

The allocation

EX-ANTE VOLATILITY	1-MONTH 5% VAR	POSITIONS	LARGEST RISK CONTR.
8.13%	3.86%	8 + BIL	17.4%
Band 7–9%	USD 3,860,465	Three baskets	VEA

TABLE 7 · CAPITAL WEIGHT AND RISK CONTRIBUTION

Capital Risk

ETF	CAPITAL	RISK	
EQUITY · 31.37% OF CAPITAL · 50.0% OF RISK			
VTI	10.90%	15.21%	
VEA	10.24%	17.42%	
IEMG	10.23%	17.36%	
BONDS · 45.07% OF CAPITAL · 25.0% OF RISK			
BND	13.33%	8.34%	
BNDX	16.48%	7.83%	
TIP	15.26%	8.84%	
REAL ASSETS · 23.55% OF CAPITAL · 25.0% OF RISK			
GLD	11.15%	15.49%	
BCI	12.40%	9.51%	
BIL	0.01%	0.00%	Cash leg

6.1 · CAPITAL VERSUS RISK

Bonds take 45% of the capital but only 25% of the risk; equity takes 31% of the capital and 50% of the risk. A dollar in equity moves the portfolio about 2.5 times as much as a dollar in bonds (volatility 15–16% against 6–7%). The capital weights therefore look defensive, but the risk is split evenly between growth and protection, by design.

6.2 · INSIDE THE BASKETS

VEA and IEMG get the same weight although IEMG is more volatile (16.3% against 15.7%), because IEMG is less correlated with VTI (0.72 against 0.82). In the whole portfolio, however, VEA and IEMG contribute more risk than VTI, because they are more correlated with bonds and gold. BNDX has the largest weight because it has the lowest volatility (5.6%). Between the two real assets, BCI gets more capital because it is less volatile; it also contributes less risk than gold (9.5% against 15.5%), because it is the least correlated asset of the portfolio.

6.3 · VOLATILITY AND VALUE AT RISK

The ex-ante volatility, $\sqrt{(w'Vw)} = 8.13\%$, is inside the band and close to the target. Under normality, the 1-month 5% VaR is $1.645 \times 8.13\% / \sqrt{12} = 3.86\%$: in one month out of twenty we expect to lose more than USD 3.86 million. We read this as a reference, not a ceiling. In the backtest the worst month of these weights was −9.4% (March 2020), and the monthly returns are skewed to the left, so the true tail is fatter than the normal one. Realised volatility can also differ from 8.1%: the window includes 2022, and in calmer years such as 2013–2021 the same weights had a volatility of 6.9%.

The backtest

We hold the SAA weights fixed and rebalance them to target at the end of every month; whatever is not invested earns the BIL return. The sample is July 2013 – September 2026 (159 months): it starts when every ETF exists, the limit being BNDX (June 2013). There are no transaction costs, since the guidelines do not mention them and monthly rebalancing only trades the drift of the weights. The measures are those of Class 2. Because the weights are estimated on the last 60 months, a single backtest would partly test the weights on the data that produced them. We therefore run four versions:

Full sample
Our weights on all 159 months. The last 60 are in-sample.

Hold-out
Our weights on the 99 months before the estimation window, which the optimiser never saw.

Split
The rule estimated once on the first 60 months, then held fixed on the following 99. No look-ahead.

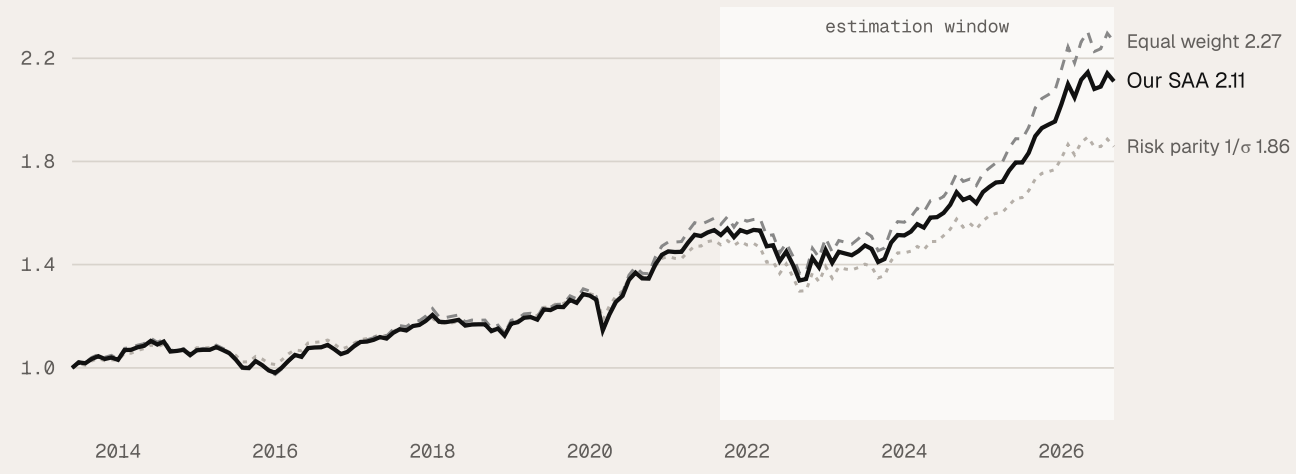
Walk-forward
The rule re-estimated every month on the previous 60 months, as an investor would have done in real time.

TABLE 8 · PERFORMANCE AND RISK OF THE SAA

	FULL SAMPLE	HOLD-OUT	SPLIT	WALK-FORWARD
Period	07.13-09.26	07.13-09.21	07.18-09.26	07.18-09.26
Return, annualised	5.92%	5.28%	9.75%	6.97%
Cumulative return	111.1%	51.5%	113.2%	71.8%
Volatility, annualised	7.39%	6.94%	10.35%	8.85%
Sharpe Ratio (t-stat.)	0.55 (2.00)	0.66 (1.91)	0.68 (1.94)	0.48 (1.37)
Sortino Ratio	0.83	1.01	1.01	0.68
Maximum drawdown	-13.04%	-11.24%	-15.57%	-14.47%
1-month 5% VaR	3.51%	3.30%	4.92%	4.20%

Sharpe Ratio on returns in excess of the 1-month T-bill; t-statistic of the mean excess return = Sharpe × √years. VaR = 1.645 × realised monthly volatility, as for the ex-ante VaR. Split and walk-forward test the rule, with weights different from ours.

FIGURE 2 · GROWTH OF USD 1, MONTHLY REBALANCING



Full sample, Jul 2013 – Sep 2026, fixed weights estimated on Oct 2021 – Sep 2026 (light band). Equal weight and risk parity 1/σ are the comparisons of Checkpoint 2, not a benchmark. [13, 14]

What the backtest tells us

TABLE 9 · CALENDAR-YEAR RETURNS

	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026*
Our SAA	**0.8**	**-5.7**	**7.3**	**11.3**	**-4.8**	**14.3**	**11.9**	**6.7**	**-9.4**	**9.0**	**8.2**	**19.3**	**7.9**
Equal weight	0.3	-6.1	7.6	13.1	-5.7	15.3	12.7	7.3	-9.6	9.8	8.9	21.6	9.1
Risk parity 1/σ	2.5	-3.8	6.3	9.1	-3.4	12.7	10.9	4.7	-10.6	8.4	6.3	15.0	5.1

*Total return in %, USD. *January – September 2026. Fixed weights of section 06 for our SAA.*

7.2 · READING THE RESULTS

Return and risk. Over the full sample, USD 1 grew to USD 2.11: 5.9% a year with 7.4% volatility and a Sharpe Ratio of 0.55. The portfolio lost money in three of the twelve full calendar years (2015, 2018, 2022), never more than 9.4%.

2022, the stress test that matters. From October 2021 to September 2022 the portfolio lost 13.0%, its maximum drawdown. In 2022 US equity fell 20% and US bonds 12.5%, so the usual hedge failed. Commodities (+15.7%) and gold (+0.8%) were the only positions that gained, which is the regime for which we hold them. The loss was recovered by March 2024. In March 2020 (−9.4% in the month, −10.9% from the peak) equities fell about 20% and commodities 17%, while bonds and gold held their value; the loss was recovered by July.

Volatility depends on the regime. With the same weights, realised volatility was 6.9% in 2013–2021, when stocks and bonds were negatively correlated, against 8.1% in the estimation window. Our weights are calibrated on the recent, less favourable regime, so they should not exceed the band if that regime persists.

The evidence is statistically weak. The t-statistics of the mean excess return are between 1.4 and 2.0: none of the tests proves at the 5% level that the SAA beats the T-bill. This is the Class 2 lesson that decades of data are needed, and it is why Checkpoint 3 discounts the backtested Sharpe Ratio.

TABLE 10 · SHARPE RATIO OF THE ELEVEN ALLOCATIONS IN THE FOUR TESTS

	FULL	HOLD-OUT	SPLIT	WALK-FWD		FULL	HOLD-OUT	SPLIT	WALK-FWD
Equal weight	0.57	0.64	0.61	0.61	Max. diversification	0.56	0.60	0.89	0.63
Risk parity 1/σ	0.49	0.78	0.33	0.42	Basket 33/33/33	0.50	0.65	0.71	0.47
Minimum variance	0.55	0.64	0.63	0.48	**Basket 50/25/25 · SAA**	**0.55**	**0.66**	**0.68**	**0.48**
Maximum Sharpe	0.78	0.64	0.60	0.66	Basket 60/20/20	0.57	0.66	0.65	0.48
ERC, 8 ETFs	0.49	0.68	0.70	0.45	Basket 70/15/15	0.58	0.66	0.62	0.49
Mean–var. at 8%	0.77	0.63	0.63	0.61					

No rule wins everywhere. Risk parity is the best on the hold-out (0.78) and the worst on the split (0.33). Maximum Sharpe leads the full sample only because its weights were fitted on the last 60 months; on the hold-out it falls to 0.64. Our basket is never the best and never the worst: this stability across samples is what we want from a portfolio that cannot be changed until Round 2, more than the highest backtested number.

The code

All the numbers in this report come from one Python notebook, attached to the report and published on our website. Running it from top to bottom reproduces every table.

`SAA_R1_our_version.ipynb`	The whole analysis. §1–2 load the prices and the T-bill and compute monthly returns; §3 describes each ETF; §4 estimates the inputs on the last 60 months; §5 defines volatility, risk contributions and VaR; §6–7 build the eleven allocations, including our basket method; §8 runs the four backtests; the checkpoint cells compute the figures of Checkpoints 1–3; §9 converts the weights into numbers of shares for Bloomberg.
`AMC_adjusted_prices_final.csv`	Bloomberg export: daily adjusted prices (dividends and splits included) of the eight ETFs, BIL and the BCOMTR index, January 2012 – October 2026.
`FRED DGS1MO`	Daily 1-month T-bill rate, downloaded by the notebook at every run and averaged by month.
`portfolio_for_PRTU.csv`	Output of §9: the table below, used to enter the portfolio in Bloomberg (PRTU).

Portfolio description

Number of shares $N_i = \lfloor w_i \times \text{USD 100 million} / P_i \rfloor$, with P_i the close of Friday 16 October 2026. Shares are rounded down, so the total never exceeds the capital; the leftover stays in cash. Every position is a single purchase at that close, entered in Bloomberg PRTU on 17–18 October and unchanged until Round 2.

SECURITY	BLOOMBERG TICKER	CCY	SHARES	PRICE	WEIGHT
Vanguard Total Stock Market ETF	`VTI US Equity`	USD	[–]	[–]	10.90%
Vanguard FTSE Developed Markets ETF	`VEA US Equity`	USD	[–]	[–]	10.24%
iShares Core MSCI Emerging Markets ETF	`IEMG US Equity`	USD	[–]	[–]	10.23%
Vanguard Total Bond Market ETF	`BND US Equity`	USD	[–]	[–]	13.33%
Vanguard Total International Bond ETF	`BNDX US Equity`	USD	[–]	[–]	16.48%
iShares TIPS Bond ETF	`TIP US Equity`	USD	[–]	[–]	15.26%
SPDR Gold Shares	`GLD US Equity`	USD	[–]	[–]	11.15%
abrdn Bloomberg All Commodity Strategy K-1 Free ETF	`BCI US Equity`	USD	[–]	[–]	12.40%
SPDR Bloomberg 1-3 Month T-Bill ETF	`BIL US Equity`	USD	[–]	[–]	0.01%
Cash, rounding leftover		USD			[–]

Price date: close of Friday 16 October 2026 for every position. Weights are the target weights of section 06; the actual weights after rounding differ by less than 0.01 percentage point.

Our allocation vs. the robust approaches

EQUAL WEIGHT

$w_i = 1 / N$

RISK PARITY

$w_i \propto 1 / \sigma_i$

Both rules are applied to our eight ETFs, fully invested. The volatilities for risk parity, the risk contributions and the volatility of each portfolio come from our own data, on the same 60 months as our covariance matrix (October 2021 – September 2026).

	OUR SAA		EQUAL WEIGHT		RISK PARITY	
ETF	WEIGHT	RISK CONTR.	WEIGHT	RISK CONTR.	WEIGHT	RISK CONTR.
VTI	10.90%	15.21%	12.50%	16.43%	8.11%	12.69%
VEA	10.24%	17.42%	12.50%	19.97%	7.89%	14.90%
IEMG	10.23%	17.36%	12.50%	20.05%	7.59%	13.98%
BND	13.33%	8.34%	12.50%	6.98%	18.29%	14.35%
BNDX	16.48%	7.83%	12.50%	5.26%	22.20%	13.47%
TIP	15.26%	8.84%	12.50%	6.45%	20.62%	14.69%
GLD	11.15%	15.49%	12.50%	15.96%	7.24%	10.73%
BCI	12.40%	9.51%	12.50%	8.91%	8.05%	5.18%
BIL	0.01%	0.00%	0.00%	0.00%	0.00%	0.00%
Equity	31.4%	50.0%	37.5%	56.4%	23.6%	41.6%
Bonds	45.1%	25.0%	37.5%	18.7%	61.1%	42.5%
Real assets	23.5%	25.0%	25.0%	24.9%	15.3%	15.9%
Volatility	8.13%		8.78%		7.23%	

WHERE THE RULES DISAGREE WITH OUR WEIGHTS

The three allocations share the same shape: every ETF holds 7–22% of the capital, all within the 7–9% band. They disagree in three places. Equity: equal weight gives it 37.5% of the capital and 56% of the risk, because it ignores that equities are 2.5 times as volatile as bonds; risk parity gives it 24% and 42%. Our 31% of capital and exactly 50% of risk lies in between, and it is the one choice in our weights that rests on a view about returns. That view is a prior that the equity premium is the most reliable premium, not an estimated mean. Bonds: risk parity puts 61% of the capital in the three bond ETFs because each looks safe on its own. But they are correlated 0.85–0.90 with each other and, since 2021, positively with equities, so 1/σ counts the same risk three times. Our ERC uses the correlations and holds 45%. Real assets: risk parity cuts gold and commodities to 15% of the capital and 16% of the risk because they are volatile on their own. It ignores that they are the least correlated assets in the universe and the only ones that gained in 2022. We keep them at 25% of the risk, on the basis of correlations and our regime analysis, not their past returns. None of our weights depends on an estimated expected return. The positions that would — 33–35% in gold under maximum Sharpe and mean–variance — are exactly the ones we rejected. Out of sample, no rule dominates: risk parity wins the hold-out (Sharpe 0.78 against our 0.66), equal weight the walk-forward (0.61 against 0.48). We therefore choose on economic grounds: we prefer the rule that gives each regime a fixed share of the risk.

Backtest protocol

1 · DESIGN

Universe	8 ETFs + BIL. BCOMTR index for BCI over the whole sample (BCI listed 2017).
Weight rule	Basket method 50/25/25, ERC inside baskets, covariance of the last 60 months.
Rebalancing	Monthly, to the SAA weights; rest in BIL; no transaction costs.
Sample	Jul 2013 – Sep 2026, 159 monthly total returns in USD, Bloomberg adjusted prices; T-bill from FRED.

2 · VALIDATION

Tuning	Oct 2021 – Sep 2026 (60 months): the window that sets the weights, as the guidelines require.
Hold-out	Jul 2013 – Sep 2021 (99 months), never used to estimate the weights. Split and walk-forward tests use only past data.
Variations	37: 11 allocations backtested in 4 designs; 23 variants compared ex ante on earlier universes; 3 changes of universe or rule.

The 3 changes were STIP → TIP, VT → VTI + VEA + IEMG, and 1/σ → ERC inside the bond basket. The hold-out is not perfectly clean: we looked at its Sharpe Ratios, with the walk-forward, when choosing among the four equity budgets, although the deciding criterion was the 8% target (Class 4: alternating between tuning and validation data contaminates the validation data).

3 · RISK

	TUNING SAMPLE	HOLD-OUT SAMPLE	FULL SAMPLE
Mean excess return, annualised (t-stat.)	3.15% (0.87)	4.61% (1.91)	4.06% (2.00)
Volatility, annualised	8.13%	6.94%	7.39%
Maximum drawdown	-13.04%	-11.24%	-13.04%
One-month 5% VaR	3.86%	3.30%	3.51%
One-month 5% VaR, USD	3,860,465	3,295,225	3,508,981
Sharpe Ratio	0.39	0.66	0.55

Fixed SAA weights. VaR = 1.645 × realised monthly volatility (normal). Walk-forward, Jul 2018 – Sep 2026: volatility 8.85%, drawdown −14.47%, VaR 4.20%, Sharpe 0.48. Cumulative returns: Figure 2.

4 · THE DISCOUNT

BACKTESTED SHARPE (HOLD-OUT)

0.66

REALISTIC (DISCOUNTED)

0.33

We start from the hold-out, the only test of our own weights on months they were not estimated on, and halve it, following the rule of thumb of Class 4. A raw 0.66 would overstate what we expect, for five reasons. (1) With 37 variations, some look good by chance alone. (2) The hold-out was consulted when we chose the budget. (3) The ETFs were chosen in 2026, knowing for example that gold rallied. (4) In 2013–2021 rates were falling and stocks and bonds hedged each other, which no longer holds. (5) There are no trading costs. The t-statistic of 1.9 also means a standard error of about 0.35 on the Sharpe Ratio. We do not discount more because our weights have a single tuned parameter, the equity budget, and use no expected returns. The value of 0.33 is consistent with the evidence that does not depend on our choices: our walk-forward gave 0.48, and in the 40-year horse race of Class 3 risk parity earned 0.53 out of sample, and a 60/40 portfolio has a Sharpe Ratio of 0.36.

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METHODOLOGY

Returns are monthly total returns in USD, before transaction costs. Round 1 is measured against its 8% volatility target; there is no common benchmark. The Sharpe Ratio is computed on returns in excess of the 1-month T-bill.

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